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Computer Based Assessment System for Reading (CBAS-R) 1.0: Hierarchy and Item Development for Comprehension

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Abstract

The purpose of this report is to define the domain of Comprehension and illustrate how this domain is assessed. Comprehension is defined as an integration of skills, including vocabulary knowledge, and its development is marked by purposeful and interactive strategies (NRP, 2003). It is the process of information gathering that is dependent upon a given text and the reader's personal background knowledge (NRP, 2003). The current report offers its own definition of the construct attempting to be measured, as well as sample items.

Keywords: comprehension, CBAS-R

Research Log

Activity	Lead Person	Timeframe
Recommendations to PI's	Ann Kroll & Lisa Krall	01/2007
Writing	Ann Kroll & Lisa Krall	01/2007

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Comprehension

The process of understanding what is heard and read. Comprehension, or constructing meaning, is the purpose of reading and listening. The NRP noted that "Comprehension has come to be viewed as the 'essence of reading' (Durkin, 1993), essential not only to academic learning but to lifelong learning as well" (NRP, 2000, p. 4-11). "Good readers have a purpose for reading" (Armbruster, p. 34), like learning how to do something, finding out new information, or for the enjoyment and entertainment that reading for pleasure brings. Good readers actively process the test "to make sense of what they read, good readers engage in a complicated process. Using their experiences and knowledge of the world, their knowledge of vocabulary and language structure, and their knowledge of reading strategies . . ., good readers make sense of the text and know how to get the most out of it. They know when they have problems with understanding" and they know "how to resolve these problems as they occur" (Armbruster, p. 48).

Comprehension occurs at different levels: literal comprehension, inferential comprehension, and evaluative or critical thinking comprehension. "Literal comprehension is concerned with the information and ideas that are explicitly stated in the" (Cooper, et al., 1979, p. 26) text. Literal comprehension uses lower order thinking skills requiring the reader to directly recall from the text. Inferential comprehension involves understanding ideas that are implied in the text, and is concerned with the skills of "drawing conclusions, generalizing, or interpreting" (Cooper et al., p. 27) the text. Evaluative comprehension, or critical thinking, "requires the reader to make an evaluative judgment" about some aspect of the text, and "integrates the reading and thinking processes" (Cooper et al., p. 29). The reader "must compare what he has read with his

values, previous experiences, or current knowledge" (Cooper et al., p. 29), or with external criteria. Both inferential and evaluative comprehension use higher order thinking skills requiring a reader to interact more with the text.

Sample Assessments

Comprehension assessments may include may include multiple choice and/or written response format asking readers to respond to questions such as:

1. Literal:
    a. What was the name of the character in the story?
    b. What happened first in the story?
2. Inferential:
    a. What do you think will happen next?
    b. Why did the character do _____?
3. Evaluative:
    a. Did the character do the right thing? Why or why not?
    b. What else could/should he have done?

Procedure

Hierarchical structure

The contents of this section provide an overview of the a priori hierarchical structure for item and CAT development. This structure will be evaluated and updated as the items are developed and field tested.

Comprehension: Item Categories

A. Category 1: Listening Comprehension
    1. Subcategory 1: Answers Literal Questions
        a. One sentence
        b. Paragraph
    2. Subcategory 2: Answers Inferential Questions
        a. One sentence
        b. Paragraph
    3. Subcategory 3: Answers Evaluative Questions
        a. One sentence
        b. Paragraph

B. Category 2: Reading Comprehension
    1. Subcategory 1: Answers Literal Questions
        a. One sentence
        b. Paragraph
    2. Subcategory 2: Answers Inferential Questions
        a. One sentence
        b. Paragraph

3. Subcategory 3: Answers Evaluative Questions
    a. One sentence
    b. Paragraph

References

Armbruster, B. (2002). *Put reading first* DIANE Publishing.

Cooper, J. D., Warncke, E. W., Ramstad, P. A., & Shipman, D. A. (1979). *The what and how of reading instruction.* Columbus, OH: Charles E. Merrill.

Durkin, D. (1993). *Teaching them to read* (6th ed.). Boston, MA: Allyn Bacon.

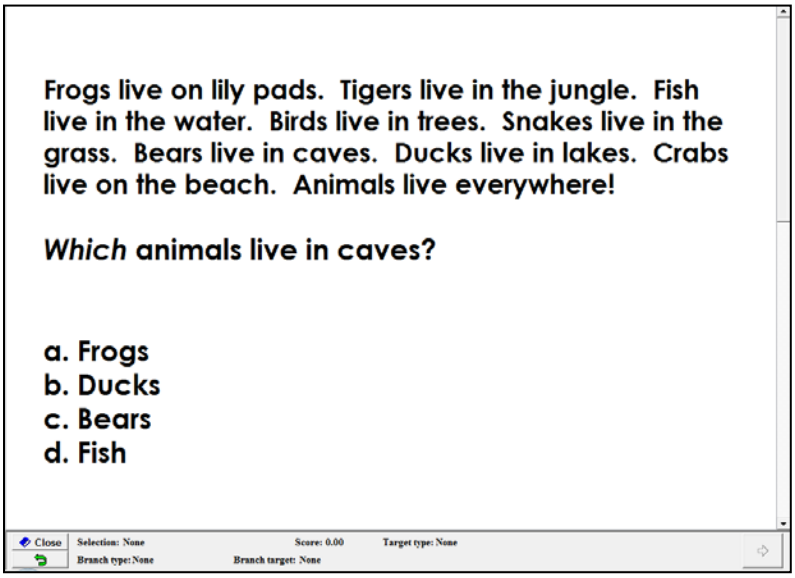
Appendix

Sample Comprehension Items

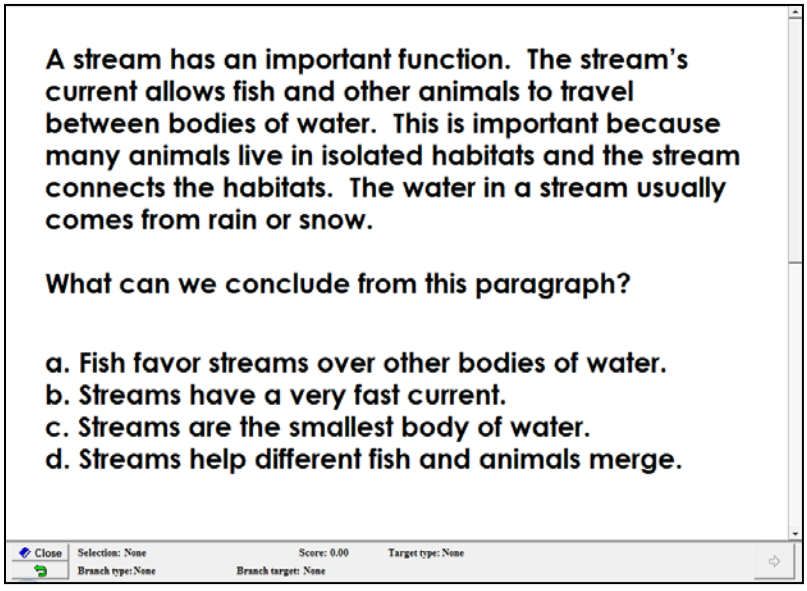
This section contains screen shots of sample test items in the hierarchical structure as defined in the Procedure section.

Literal Comprehension

Audio: “Choose the best answer.”

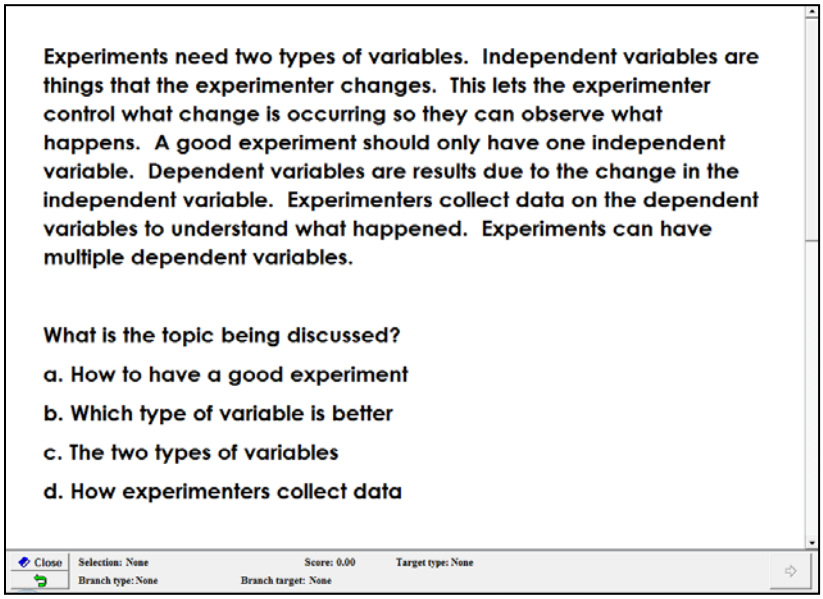


Audio: “Choose the best answer.”

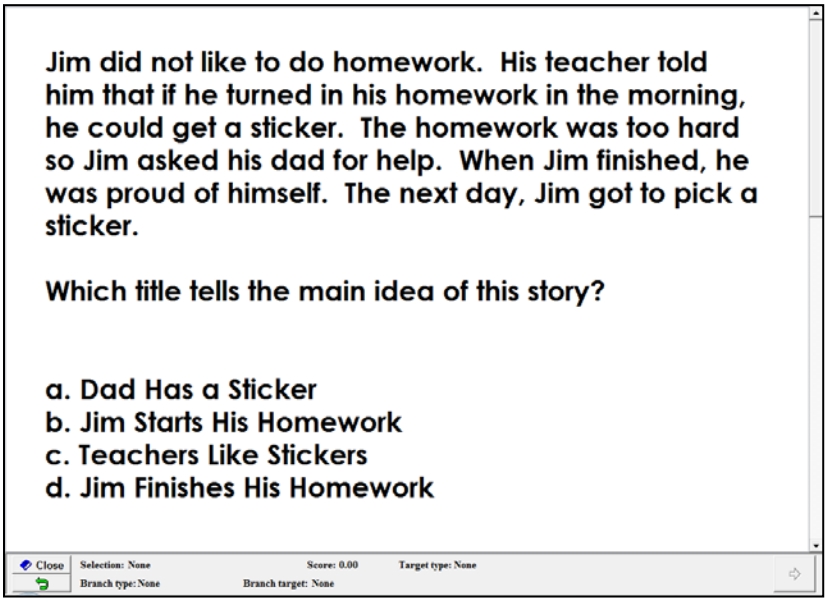


Expository Comprehension

Audio: “Choose the best answer.”

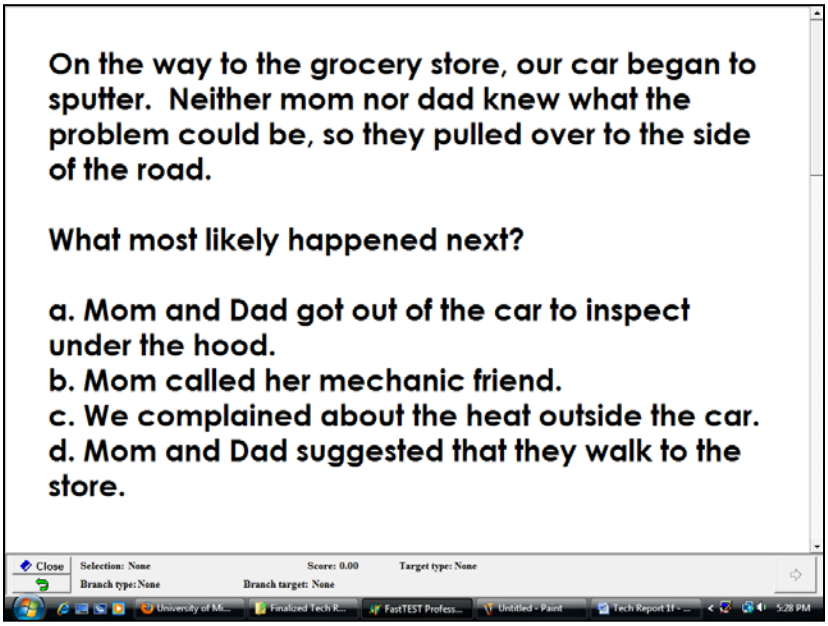


Audio: “Choose the best answer.”



Inferential Comprehension

Audio: “Choose the best answer.”



Audio: “Choose the best answer.”

